Hakan Alex Kucukhuseyin

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EDUCATION

Drexel University Philadelphia, PA
B.S. Electrical Engineering & Computer Engineering (Double Major), Minor in Finance *Expected June 2028*
GPA: 3.6 | **Finance Minor GPA:** 4.0
Honors: Dean's List (September 2024 – Present) | 1st Place, Drexel Freshman Engineering Design (2025)
Relevant Coursework: Data Structures, Circuits II, Signals & Systems, Microcontrollers, Probability & Data Analytics

EXPERIENCE

Blues Yachting Mugla, Turkey
Software Developer *June 2023 – June 2024 | June 2025 – August 2025*
- Built a React/Node.js/PostgreSQL operations platform that replaced spreadsheet workflows across quoting, CRM, inventory, and payments with 18 integrated modules.
- Cut quote preparation from 15–20 minutes to a few minutes (∼85%) through automated Turkish tax calculations, multi-line pricing, and one-click PDF generation.
- Deployed with Docker and JWT authentication; set up GitHub Actions CI that runs lint, unit, and end-to-end tests against a disposable PostgreSQL database on every push.
- Partnered with sales, service, and operations staff to gather requirements, prioritize workflows, train users, and iterate on product feedback.

PROJECTS

VitalView – Non-Invasive Glucose Monitor | Demo: hakanalex-k.github.io/vitalview Philadelphia, PA
Team Lead, Freshman Engineering Design (1st Place) *March 2025 – June 2025*
- Led an end-to-end prototype combining ESP32/PPG sensing, a Node.js API, TensorFlow/Keras inference, and a React dashboard for glucose and heart-rate monitoring.
- Trained a model on 600-feature IR/red signal vectors; added checks to reject low-amplitude or flatline readings.

Turkish University Rankings Platform Philadelphia, PA
Sole Developer, Personal Project *January 2025 – Present*
- Designed and built a platform that turns fragmented YÖKAtlas admissions data into searchable, comparable university and program information for prospective students.
- Built a Python ETL pipeline that loads YÖKAtlas data (81 provinces, 5 score types, 4 admission years) into PostgreSQL.
- Reconciled ∼500 universities via normalization, transliteration, and fuzzy matching; built a TypeScript/Express API.

Hack Computer – CPU & Memory System (Verilog) Philadelphia, PA
Course Project, ECE 350 *June 2026 – August 2026*
- Built a stored-program Hack computer in Verilog: ALU, registers, program counter, control logic, and RAM/ROM.
- Implemented the datapath and memory-mapped I/O for RAM, screen, and keyboard; tested it with Hack programs.

Drexel Turkish Student Association Website | Live site: drexeltsa.vercel.app *October 2026*
- Built and deployed a full-stack Next.js/React/TypeScript site (Vercel, Neon Postgres, $0/month) with an admin dashboard where board members publish events, blog posts, photo galleries, and team profiles.
- Secured the site with scrypt password hashing, hashed server-side sessions, CSRF origin checks, a nonce-based Content Security Policy, and database-backed rate limiting.

TECHNICAL SKILLS

Languages C, Python, JavaScript/TypeScript, SQL, Verilog
Software React, Node.js, Express, TensorFlow/Keras, PostgreSQL, Docker, Git, GitHub Actions, Linux, REST APIs
Hardware ESP32, microcontrollers, digital logic, I2C, computer architecture

ACTIVITIES

IEEE, Drexel Chapter – Member September 2024 – Present
Drexel Turkish Student Association – Event Coordinator